



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 15.11.2022

General Instructions:

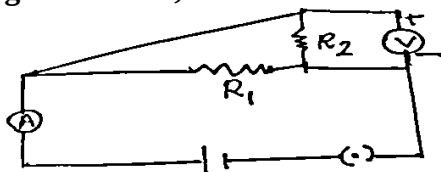
- This paper consists of three printed pages.
- All questions are compulsory.
- Marks will be deducted for spelling errors.

SECTION A
(Attempt all questions.)

Question 1

Choose the correct answers to the questions from the given options. [1 × 7 = 7]

(i) In the given circuit, name the circuit components that are connected in parallel.



- a) R_1 and R_2 b) R_1 , R_2 and V c) R_2 and V d) R_1 and V

(ii) If R_1 and R_2 be the resistance of the filament of 40 W and 60 W respectively operating at 220 V, then

- a) $R_1 < R_2$ b) $R_2 < R_1$ c) $R_1 = R_2$ d) $R_1 \geq R_2$

(iii) A boy records that 4000 joule of work is required to transfer 10 coulomb of charge between two points of a resistor of resistance 50 Ω . The current passing through it is

- a) 2A b) 4 A c) 8 A d) 16 A

(iv) Which one of the following statements is correct?

- a) Live wire has zero potential.
b) Fuse is connected in a neutral wire.
c) Potential of live and earth wire is always the same.
d) Potential of neutral and earth wire is same.

(v) The main switch breaks the connection of.

- a) live wire b) neutral wire c) Both (a) & (b) d) earth wire

(vi) The heating element of an electric iron is made up of

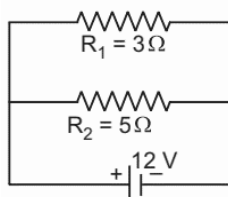
- a) copper b) nichrome c) aluminium d) iron

(vii) A coil in the heater consumes power P on passing current. If it is cut into halves and joined in parallel, it will consume power

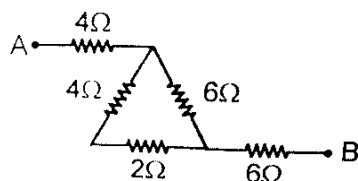
- a) P b) $P/2$ c) $2P$ d) $4P$

Question 2

- (i) a) What do you mean by the statement, “The resistivity of a body is $4.9 \times 10^{-7} \Omega\text{m}$ ”?
 b) A man has 10 resistances of 5Ω each. What is the smallest resistance that can be obtained by combining them?
 c) Mention any one factor that affects the emf of a cell. [3]
 (ii) State two differences between the emf and terminal voltage of a cell. [2]
 (iii) Why is it not advisable to use a piece of copper wire as fuse in an electric circuit? [2]
 (iv) In the following circuit find the electrical energy spent in R_1 for one minute. [2]



- (v) Mention color of the wire by which the following parts of an electric appliances are connected a) fuse b) metallic case. [2]
 (vi) Calculate the resistance across AB. [2]



SECTION B

(Attempt all questions.)

Question 3

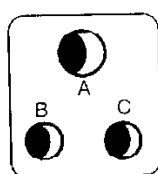
[3 + 3 + 4 = 10]

- i) It is given that the power rating of an electrical appliance is 200W- 220 V. Now if the appliance is connected to a 110 V power supply, then find the Power consumption? Why is the voltage raised in the power generating station before its transmission?
 ii) a) What are super conductors?
 b) Name a substance whose resistance decrease with increase in temperature.
 c) What is the necessary condition for a conductor to obey Ohm’s law?
 iii) Why is a switch always connected to live wire? A fuse is rated as 6A. State whether it can be used with an electric appliance of rating 2 kW, 250 V. Justify your answer.

Question 4

[3 + 3 + 4 = 10]

- i) The following figure shows the connecting points of a socket outlet. State the connections which are made to the holes marked A, B and C. Why does A have different size than the rest?



- ii) A metallic wire of resistance $30\ \Omega$ is tripled on itself. How will the resistance and resistivity change?
- iii) The circuit diagram in the given figure shows three resistors $2\ \Omega$, $4\ \Omega$ and $R\ \Omega$ connected to a battery of e.m.f. 2V and internal resistance $3\ \Omega$. If main current of 0.25 A flows through the circuit, Find
- the value of R
 - the potential difference across $2\ \Omega$ resistor
 - the potential difference across the terminals of the cell.

